

Lesson 50 Practice: Answers and Rationales

JKSSB Computer Awareness

October 10, 2026

PYQ provenance: Q1–Q4 adapt Computer Instructor/Operator 2025 Q1–Q4; Q5 adapts Website Operator 2026 Q73. These are paraphrases, not verbatim reproductions. The Website Operator answer key is provisional.

1. **D.** The ASCII decimal code for \$ is 36. 32 is the space code; 30 and 34 are other control/punctuation values, not the dollar sign.
2. **C.** The integer bits give $128 + 64 + 16 + 4 + 1 = 213$; the fractional bits give $0/2 + 1/4 = 0.25$, so the value is 213.25. A, B, and D do not match the positional sum.
3. **B.** UTF-8 is variable-length and preserves ASCII values in the ASCII range. A falsely says fixed-length; C wrongly limits Unicode to Latin; D incorrectly says every character uses eight bytes.
4. **A.** $278 = 256 + 16 + 4 + 2$, which sets bits 8, 4, 2, and 1 to give 100010110_2 . B, C, and D encode different decimal values.
5. **A.** $1010_2 = 8 + 2 = 10$. B is 9, C is 12, and D is 14.
6. **B.** Octal is base 8. A is binary, C decimal, and D hexadecimal.
7. **D.** Octal digits range from 0 through 7, so 8 is not allowed. 0, 5, and 7 are valid octal digits.
8. **C.** $2F_{16} = 2(16) + 15 = 47$. A is 31, B is 32, and D is 63; none equals the place-value sum.
9. **A.** Grouping from the right gives $001\ 111\ 111_2$, or $1\ 7\ 7_8$. B has the wrong base label; C drops a leading group; D gives the wrong base label.
10. **B.** BCD encodes each decimal digit separately: 2 is 0010 and 5 is 0101. A is the ordinary binary value 25; C reverses the digit nibbles; D is not the two-digit BCD form.
11. **D.** BCD is a decimal-digit code, not another name for Unicode. A correctly describes original ASCII; B correctly describes Unicode; C correctly describes UTF-8.
12. **A.** Unicode assigns code points; UTF-8 is one encoding that represents those code points with variable-length byte sequences. B conflates UTF-8 and BCD; C confuses Unicode with a four-bit decimal code; D falsely claims a fixed eight-byte length.
13. **A.** The channels are red, green, and blue, with 8 bits each in this common 24-bit representation. B, C, and D are not the RGB channel names.
14. **B.** XOR outputs 1 when inputs differ, and 1 and 0 differ. A is the AND result; C is not a Boolean output; D incorrectly makes a combinational gate depend on a clock.

15. **C.** AND is 1 only when both inputs are 1. OR is 1 if either input is 1; XOR is 1 when they differ; NOT takes one input.
16. **A.** NAND is the negation of AND; with inputs 1 and 1, AND is 1 and NAND is 0. B is the AND result; C is not a Boolean value; D is defined for these inputs.
17. **A.** NOT has one input and returns its complement. B describes AND; C describes XOR; D confuses a logic gate with colour channels.
18. **B.** I and II are true: three 8-bit RGB channels total 24 bits. III is false because formats can use other channels and bit depths. A omits II; C includes false III and omits I; D includes false III.
19. **C.** $1011_2 = 8 + 2 + 1 = 11$. A is 9, B is 10, and D is 13.
20. **C.** One hexadecimal digit represents four binary bits. Two bits correspond to one octal digit; three bits form an octal group; eight bits make a byte, not a hex digit.
21. **A.** For inputs 0,0, AND=0, OR=0, XOR=0, and NAND=1. B assigns an incorrect AND and NAND; C gives incorrect OR and XOR; D gives incorrect AND, OR, and XOR.
22. **A.** 1010 is $8+2 = 10$. B, C, and D miscalculate the positional values.
23. **A.** Three bits give $2^3 = 8$ values. B, C, and D confuse bits with value counts.
24. **A.** AND needs all inputs at 1. B needs any input; C inverts one input; D is not a gate.
25. **A, B, C.** The three conversions are exact. D misreads 101 as 6.
26. **A, B, C.** The AND, OR, and NOT behaviours are correct. D contradicts the AND rule.
- Q27. B.** Binary 1010 equals $8+2=10$ in decimal. The other options do not match its place-value sum.
- Q28. C.** Hexadecimal digits A through F represent decimal 10 through 15.
- Q29. A.** Binary-coded decimal encodes each decimal digit separately in four bits.
- Q30. A.** XOR is 1 when inputs differ. AND requires both inputs 1; OR requires at least one; NOT has one input.
- Q31. A, B, C.** The first three are correct. Binary 1110 equals decimal 14; decimal 10 is binary 1010.